

Lab - Exploring DNS Traffic (Instructor Version)

Instructor Note: Red font color or gray highlights indicate text that appears in the instructor copy only.

Objectives

Part 1: Capture DNS Traffic

Part 2: Explore DNS Query Traffic

Part 3: Explore DNS Response Traffic

Background / Scenario

Wireshark is an open source packet capture and analysis tool. Wireshark gives a detailed breakdown of the network protocol stack. Wireshark allows you to filter traffic for network troubleshooting, investigate security issues, and analyze network protocols. Because Wireshark allows you to view the packet details, it can be used as a reconnaissance tool for an attacker.

In this lab, you will install Wireshark and use Wireshark to filter for DNS packets and view the details of both DNS query and response packets.

Required Resources

- 1 PC with internet access and Wireshark installed

Instructor Note: Using a packet sniffer such as Wireshark may be considered a breach of the security policy of the school. It is recommended that permission is obtained before running Wireshark for this lab. If using a packet sniffer such as Wireshark is an issue, the instructor may wish to assign the lab as homework or perform a walk-through demonstration.

Instructions

Part 1: Capture DNS Traffic

Step 1: Download and install Wireshark.

- Download the latest stable version of Wireshark from www.wireshark.org. Choose the software version you need based on your PC's architecture and operating system.
- Follow the on-screen instructions to install Wireshark. If you are prompted to install USBPcap, **do NOT** install USBPcap for normal traffic capture. USBPcap is experimental, and it could cause USB problems on your PC.

Step 2: Capture DNS traffic.

- Start Wireshark. Select an active interface with traffic for packet capture.
- Clear the DNS cache.
 - 1) In Windows, enter **ipconfig /flushdns** in Command Prompt.
 - 2) For the majority of Linux distributions, one of the following utilities is used for DNS caching: Systemd - Resolved, DNSMasq, and NSCD. If your Linux distribution does not use one of the listed utilities, please perform an internet search for the DNS caching utility for your Linux distribution.
 - (i) Identify the utility used in your Linux distribution by checking the status:

Systemd-Resolved: **systemctl status systemd-resolved.service**

Dnsmasq: **systemctl status dnsmasq.service**

NSCD: **systemctl status nscd.service**

- (ii) If you are using systemd-resolved, enter **systemd-resolve --flush-caches** to flush the cache for Systemd-Resolved before restarting the service. The following commands restart the associated service using elevated privileges:

Systemd-Resolved: **sudo systemctl restart systemd-resolved.service**

Dnsmasq: **sudo systemctl restart dnsmasq.service**

NSCD: **sudo systemctl restart nscd.service**

- 3) For the macOS, enter **sudo killall -HUP mDNSResponder** to clear the DNS cache in the Terminal. Perform an internet search for the commands to clear the DNS cache for an older OS.

- At a command prompt or terminal, type **nslookup** enter the interactive mode.
- Enter the domain name of a website. The domain name www.cisco.com is used in this example.
- Type **exit** when finished. Close the command prompt.
- Click **Stop capturing packets** to stop the Wireshark capture.

Part 2: Explore DNS Query Traffic

- Observe the traffic captured in the Wireshark Packet List pane. Enter **udp.port == 53** in the filter box and click the arrow (or press enter) to display only DNS packets. **Note:** The provided screenshots are just examples. Your output may be slightly different.

No.	Time	Source	Destination	Protocol	Length	Info
1586	44.586362000	192.168.1.7	192.168.1.1	DNS	73	Standard query 0x0002 A www.cisco.com
1587	44.759772000	192.168.1.1	192.168.1.7	DNS	525	Standard query response 0x0002 A www.cisco.com
1588	44.765654800	192.168.1.7	192.168.1.1	DNS	73	Standard query 0x0003 AAAA www.cisco.com
1589	44.963934300	192.168.1.1	192.168.1.7	DNS	537	Standard query response 0x0003 AAAA www.cisco.com

Frame 1586:	Packet, 73 bytes on wire (584 bits), 73 bytes captured (584 bits) on interface \Device\NPF_{B84AE000-0010-0000-0000-000000000000}	64	6d
Ethernet II, Src: Intel_14:ca:e4 (00:42:38:14:ca:e4), Dst: HuaweiTechno_1f:fa:44 (64:6d:6c:1f:fa:44)	0010	00	3b
Internet Protocol Version 4, Src: 192.168.1.7, Dst: 192.168.1.1	0020	01	01
User Datagram Protocol, Src Port: 50463, Dst Port: 53	0030	00	00
Domain Name System (query)	0040	03	63

- Select the DNS packet contains **Standard query** and **A www.cisco.com** in the Info column.
- In the Packet Details pane, notice this packet has Ethernet II, Internet Protocol Version 4, User Datagram Protocol and Domain Name System (query).

- d. Expand **Ethernet II** to view the details. Observe the source and destination fields.

udp.port == 53						
No.	Time	Source	Destination	Protocol	Length	Info
1270	37.980191400	192.168.1.1	192.168.1.7	DNS	296	Standard query response 0xcb39 HTTPS euc-common.online.of
1271	37.988807600	192.168.1.1	192.168.1.7	DNS	407	Standard query response 0x7785 A euc-common.online.office
1384	39.884102800	192.168.1.7	192.168.1.1	DNS	98	Standard query 0xf916 HTTPS euc-word-telemetry.officeapps
1385	39.884337700	192.168.1.7	192.168.1.1	DNS	98	Standard query 0x1a2c A euc-word-telemetry.officeapps.live
1387	40.134576200	192.168.1.1	192.168.1.7	DNS	266	Standard query response 0xf916 HTTPS euc-word-telemetry.o
1389	40.134584900	192.168.1.1	192.168.1.7	DNS	447	Standard query response 0x1a2c A euc-word-telemetry.office
1438	40.882845600	192.168.1.7	192.168.1.1	DNS	93	Standard query 0x8dfb HTTPS euc-word-edit.officeapps.live
1439	40.883325100	192.168.1.7	192.168.1.1	DNS	93	Standard query 0xd7f4 A euc-word-edit.officeapps.live.com
1445	40.989120200	192.168.1.1	192.168.1.7	DNS	301	Standard query response 0x8dfb HTTPS euc-word-edit.office
1447	40.998542000	192.168.1.1	192.168.1.7	DNS	412	Standard query response 0xd7f4 A euc-word-edit.officeapps
1586	44.586362000	192.168.1.7	192.168.1.1	DNS	73	Standard query 0x0002 A www.cisco.com
1587	44.759772000	192.168.1.1	192.168.1.7	DNS	525	Standard query response 0x0002 A www.cisco.com CNAME www.c
1588	44.765654800	192.168.1.7	192.168.1.1	DNS	73	Standard query 0x0003 AAAA www.cisco.com
1589	44.963934300	192.168.1.1	192.168.1.7	DNS	537	Standard query response 0x0003 AAAA www.cisco.com CNAME ww
1755	54.760184700	192.168.1.7	192.168.1.1	DNS	96	Standard query 0xb79e HTTPS sdmntpritalynorth.oaiuserconte
1756	54.760331500	192.168.1.7	192.168.1.1	DNS	96	Standard query 0x6116 A sdmntpritalynorth.oaiusercontent.c
1762	54.808644900	192.168.1.1	192.168.1.7	DNS	154	Standard query response 0xb79e HTTPS sdmntpritalynorth.oa
1763	54.820872800	192.168.1.1	192.168.1.7	DNS	445	Standard query response 0x6116 A sdmntpritalynorth.oaiuser
1904	55.872333100	192.168.1.7	192.168.1.1	DNS	83	Standard query 0x007a HTTPS spclient.wg.spotify.com

What are the source and destination MAC addresses? Which network interfaces are these MAC addresses associated with?

The source MAC address is 00:42:38:14:ca:e4, associated with the PC's NIC. The destination MAC address is 64:6d:6c:1f:fa:44, associated with the default gateway.

- e. Expand **Internet Protocol Version 4**. Observe the source and destination IPv4 addresses.

Time	Source	Destination	Protocol	Length	Info
86 44.586362000	192.168.1.7	192.168.1.1	DNS	73	Standard query 0x0002 A www.cisco.com
87 44.759772000	192.168.1.1	192.168.1.7	DNS	525	Standard query response 0x0002 A www.cisco.com CNAME www.c
88 44.765654800	192.168.1.7	192.168.1.1	DNS	73	Standard query 0x0003 AAAA www.cisco.com
89 44.963934300	192.168.1.1	192.168.1.7	DNS	537	Standard query response 0x0003 AAAA www.cisco.com CNAME www

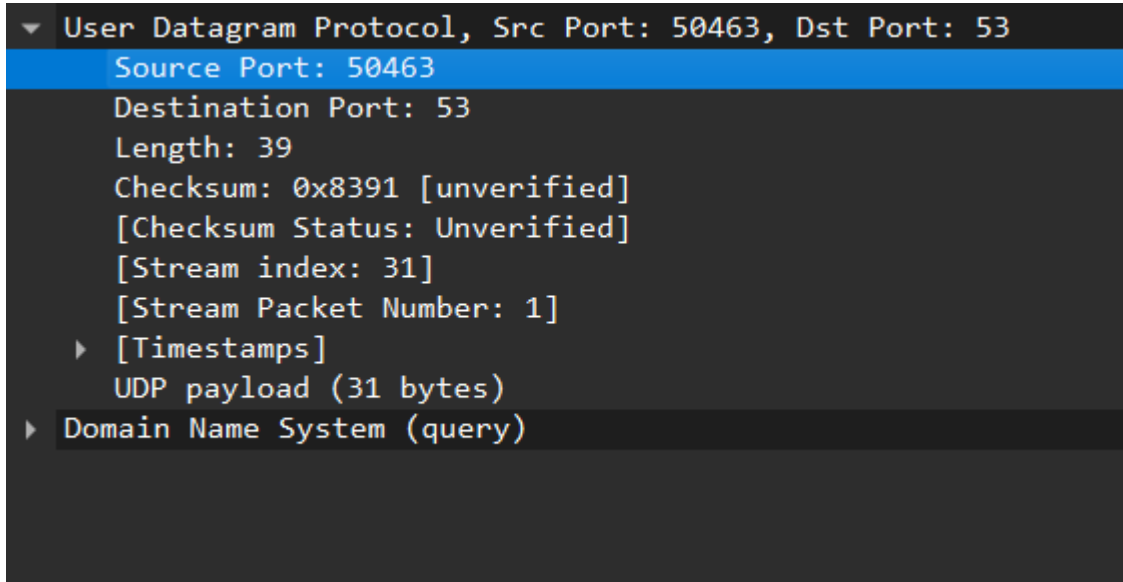
```

Frame 1586: Packet, 73 bytes on wire (584 bits), 73 bytes captured (584 bits) on interface \Device\NPF_{B84...}
  Ethernet II, Src: Intel_14:ca:e4 (00:42:38:14:ca:e4), Dst: HuaweiTechno_1f:fa:44 (64:6d:6c:1f:fa:44)
    Destination: HuaweiTechno_1f:fa:44 (64:6d:6c:1f:fa:44)
    Source: Intel_14:ca:e4 (00:42:38:14:ca:e4)
    Type: IPv4 (0x0800)
    [Stream index: 0]
  Internet Protocol Version 4, Src: 192.168.1.7, Dst: 192.168.1.1
    User Datagram Protocol, Src Port: 50463, Dst Port: 53
      Source Port: 50463
      Destination Port: 53
      Length: 39
      Checksum: 0x8391 [unverified]
      [Checksum Status: Unverified]
      [Stream index: 31]
      [Stream Packet Number: 1]
      [Timestamps]
      UDP payload (31 bytes)
    Domain Name System (query)
      Transaction ID: 0x0002
      Flags: 0x0100 Standard query
        0... .. = Response: Message is a query
        .000 0... .. = Opcode: Standard query (0)
        .... ..0. .... = Truncated: Message is not truncated
        .... ..1 .... = Recursion desired: Do query recursively
        .... ..0.. .... = Z: reserved (0)
        .... ....0 .... = Non-authenticated data: Unacceptable
      Questions: 1
      Answer RRs: 0
      Authority RRs: 0
      Additional RRs: 0
  
```

What are the source and destination IP addresses? Which network interfaces are these IP addresses associated with?

The source IP address is 192.168.1.7, associated with the PC is network interface The destination IP address is 192.168.1.1, associated with the DNS server/default gateway

- f. Expand the **User Datagram Protocol**. Observe the source and destination ports.



What are the source and destination ports? What is the default DNS port number?

- **e. The source port is 50463 and the destination port is 53. The default DNS port number is 53**

- g. Determine the IP and MAC address of the PC.

- 1) In a Windows command prompt, enter **arp -a** and **ipconfig /all** to record the MAC and IP addresses of the PC.
- 2) For Linux and macOS PC, enter **ifconfig** or **ip address** in a terminal.

Compare the MAC and IP addresses in the Wireshark results to the IP and MAC addresses. What is your observation?

- h. The IP and MAC addresses captured in the Wireshark results are the same**
 - i. as the addresses listed in the ipconfig /all command.**

- j. Expand **Domain Name System (query)** in the Packet Details pane. Then expand the **Flags** and **Queries**.

- k. Observe the results. The flag is set to do the query recursively to query for the IP address to www.cisco.com.

```

UDP payload (31 bytes)
Domain Name System (query)
Transaction ID: 0x0002
▼ Flags: 0x0100 Standard query
  0... .. = Response: Message is a query
  .000 0... .. = Opcode: Standard query (0)
  .... ..0. .... = Truncated: Message is not truncated
  .... ..1 .... = Recursion desired: Do query recursively
  .... ..0... .. = Z: reserved (0)
  .... ..0 .... = Non-authenticated data: Unacceptable
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 0
▼ Queries
  ► www.cisco.com: type A, class IN
  [Response In: 1587]
  
```

Part 3: Explore DNS Response Traffic

- a. Select the corresponding response DNS packet has **Standard query response** and **A www.cisco.com** in the Info column.

Time	Source	Destination	Protocol	Length	Info
5 44.586362000	192.168.1.7	192.168.1.1	DNS	73	Standard query 0x0002 A www.cisco.com
7 44.759772000	192.168.1.1	192.168.1.7	DNS	525	Standard query response 0x0002 A www.cisco.com
8 44.765654800	192.168.1.7	192.168.1.1	DNS	73	Standard query 0x0003 AAAA www.cisco.com
9 44.963934300	192.168.1.1	192.168.1.7	DNS	537	Standard query response 0x0003 AAAA www.cisco.com

Frame 1587: Packet, 525 bytes on wire (4200 bits), 525 bytes captured on interface 0	0000	00 42 38 14 ca e4 64 6d 6c 1f fa 44 08 00 42
Ethernet II, Src: HuaweiTechno_1f:fa:44 (64:6d:6c:1f:fa:44), Dst: Intel_14:ca:e4 (00:42:38:14:ca:e4)	0010	01 ff 00 00 40 00 40 11 b5 95 c0 a8 01 01 c0
▼ Destination: Intel_14:ca:e4 (00:42:38:14:ca:e4)	0020	01 07 00 35 c5 1f 01 eb 7c b0 00 02 81 80 00
.... ..0. = LG bit: Globally unique address	0030	00 04 00 08 00 09 03 77 77 77 05 63 69 73 61
.... ..0. = IG bit: Individual address (unicast)	0040	03 63 6f 6d 00 00 01 00 01 c0 0c 00 05 00 00
▼ Source: HuaweiTechno_1f:fa:44 (64:6d:6c:1f:fa:44)	0050	00 0d 6b 00 1a 03 77 77 77 05 63 69 73 63 61
.... ..0. = LG bit: Globally unique address	0060	63 6f 6d 06 61 6b 61 64 6e 73 03 6e 65 74 00
.... ..0. = IG bit: Individual address (unicast)	0070	2b 00 05 00 01 00 00 00 c3 00 1a 05 77 77 77
Type: IPv4 (0x0800)	0080	73 05 63 69 73 63 6f 03 63 6f 6d 07 65 64 61
[Stream index: 0]	0090	6b 65 79 c0 40 c0 51 00 05 00 01 00 00 18 64
Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.7	00a0	18 05 65 32 38 36 37 04 64 73 63 61 0a 61 61
User Datagram Protocol, Src Port: 53, Dst Port: 50463	00b0	6d 61 69 65 64 67 65 c0 40 c0 77 00 01 00 00
Domain Name System (response)	00c0	00 00 14 00 04 b8 19 34 77 c0 7d 00 02 00 00
	00d0	00 0d 2b 00 09 06 6e 32 64 73 63 61 c0 82 c0
	00e0	00 02 00 01 00 00 0d 2b 00 09 06 6e 33 64 73

- b. Destination MAC: 00:42:38:14:ca:e4
 Source IP: 192.168.1.1
 Destination IP: 192.168.1.7
 Source Port: 53
 Destination Port: 50463

- c. Expand **Domain Name System (response)**. Then expand the **Flags**, **Queries**, and **Answers**.
 d. Observe the results.

Can the DNS server do recursive queries?

Yes, the DNS server can handle recursive queries.

Time	Source	Destination	Protocol	Length	Info
86.44.586362000	192.168.1.7	192.168.1.1	DNS	73	Standard query 0x0002 A www.cisco.com
87.44.759772000	192.168.1.1	192.168.1.7	DNS	525	Standard query response 0x0002 A www.cisco.com CNAME www.cisco.com
88.44.765654800	192.168.1.7	192.168.1.1	DNS	73	Standard query 0x0003 AAAA www.cisco.com
89.44.963934300	192.168.1.1	192.168.1.7	DNS	537	Standard query response 0x0003 AAAA www.cisco.com CNAME www.cisco.com

Frame 1587: Packet, 525 bytes on wire (4200 bits), 525 bytes captured on interface 0 Ethernet II, Src: HuaweiTechno_1f:fa:44 (64:6d:6c:1f:fa:44), Dst: 00:42:38:14:ca:e4 Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.7 User Datagram Protocol, Src Port: 53, Dst Port: 50463 Domain Name System (response) Transaction ID: 0x0002 Flags: 0x8180 Standard query response, No error 1... .. = Response: Message is a response 0000... .. = Opcode: Standard query (0) = Authoritative: Server is not an authoritative source = Truncated: Message is not truncated = Recursion desired: Do query recursively = Recursion available: Server can do recursive queries = Z: reserved (0) = Answer authenticated: Answer/authority = Non-authenticated data: Unacceptable = Reply code: No error (0) Questions: 1 Answer RRs: 4 Authority RRs: 8 Additional RRs: 9 Queries www.cisco.com: type A, class IN Name: www.cisco.com [Name Length: 13] [Label Count: 3] Type: A (1) (Host Address) Class: IN (0x0001)	0000 00 42 38 14 ca e4 6d 6c 1f fa 44 08 00 45 00 B8 .. dm l.. 0010 01 ff 00 00 40 00 40 11 b5 95 c0 a8 01 01 c0 a8 ... @ @ .. 0020 01 07 00 35 c5 1f 01 eb 7c b0 00 02 81 80 00 01 ... 5 0030 00 04 00 08 00 09 03 77 77 05 63 69 73 63 6f w ww 0040 03 63 6f 6d 00 00 01 00 01 c0 0c 00 05 00 01 00 .com..... 0050 00 0d 6b 00 1a 03 77 77 77 05 63 69 73 63 6f 03 ..k...ww w.c 0060 63 6f 6d 06 61 6b 61 64 6e 73 03 6e 65 74 00 c0 com.akad ns 0070 2b 00 05 00 01 00 00 00 c3 00 1a 05 77 77 77 64 +..... .. 0080 73 05 63 69 73 63 6f 03 63 6f 6d 07 65 64 67 65 s-cisco. cor 0090 6b 65 79 c0 40 c0 51 00 05 00 01 00 00 18 64 00 key.@.Q. ... 00a0 18 05 65 32 38 36 37 04 64 73 63 61 0a 61 6b 61 ..e2867. dsc 00b0 6d 61 69 65 64 67 65 c0 40 c0 77 00 01 00 01 00 maiedge. @.v 00c0 00 00 14 00 04 b8 19 34 77 c0 7d 00 02 00 01 004 w. 00d0 00 0d 2b 00 09 06 6e 32 64 73 63 61 c0 82 c0 7d ++...n2 dsc 00e0 00 02 00 01 00 00 0d 2b 00 09 06 6e 33 64 73 63+ .. 00f0 61 c0 82 c0 7d 00 02 00 01 00 00 0d 2b 00 09 06 a...}... .. 0100 6e 30 64 73 63 61 c0 82 c0 7d 00 02 00 01 00 00 n0dsca... }.. 0110 0d 2b 00 09 06 6e 37 64 73 63 61 c0 82 c0 7d 00 ++...n7d sca 0120 02 00 01 00 00 0d 2b 00 09 06 6e 35 64 73 63 61+... .. 0130 c0 82 c0 7d 00 02 00 01 00 00 0d 2b 00 09 06 6e ...}..... .. 0140 36 64 73 63 61 c0 82 c0 7d 00 02 00 01 00 00 0d 6dsca... }.. 0150 2b 00 09 06 6e 34 64 73 63 61 c0 82 c0 7d 00 02 +...n4ds ca 0160 00 01 00 00 0d 2b 00 09 06 6e 31 64 73 63 61 c0+... .. 0170 82 c0 ea 00 01 00 01 00 00 0d 2b 00 04 52 66 b4+... .. 0180 b5 c1 14 00 01 00 01 00 00 0d 2b 00 04 4d 89 b2 0190 0e c0 ab 00 01 00 01 00 00 0d 2b 00 04 52 66 b4 01a0 b6 c1 3e 00 01 00 01 00 00 0d 2b 00 04 4d 89 b2 01b0 3e c0 d5 00 01 00 01 00 00 0d 2b 00 04 58 dd 51 01c0 c0 c1 29 00 01 00 01 00 00 0d 2b 00 04 4d 89 b2 01d0 44 c0 c0 00 01 00 01 00 00 0d 2b 00 04 4d 89 b2
---	---

- e. Observe the CNAME and A records in the Answers details.

How do the results compare to nslookup results?

The results in Wireshark should be the same as the results from nslookup in the Command Prompt or terminal.

Reflection

1. From the Wireshark results, what else can you learn about the network when you remove the filter?

Without the filters, the results display other packets, such as DHCP and ARP. From these packets and the information contained within them, you can learn about other devices and their functions within the LAN.

2. How can an attacker use Wireshark to compromise your network security?

An attacker on the LAN can use Wireshark to observe the network traffic and can get sensitive information in the packet details if the traffic is not encrypted.